

Secure Diagnostic Laboratory Network

Project Portfolio Summary
Cisco Packet Tracer Network Security Simulation

Project	Designing a Secure Network for Diagnostic Data Transmission at AVM Labs
Project type	Group diploma project
Individual role	Network Implementer & Simulation Lead
Primary tools	Cisco Packet Tracer · Wireshark · Python
Project period	February 2026 – June 2026

Portfolio note: This document is a concise public-facing summary of the academic project. It is not a copy of the original college report and does not include student IDs, signatures, or other personal submission material.

Prepared for educational and cybersecurity portfolio purposes

1. Project Overview

The project involved designing and simulating a secure network for AVM Labs, a diagnostic laboratory in Qurum, Muscat. The scenario involved the laboratory exchanging diagnostic results with 17 associated clinic branches. The project identified unsecured report transmission and a lack of adequate network security controls as the main problems to address.

The proposed solution was implemented as a Cisco Packet Tracer simulation. The network was segmented into logical areas and protected using access-control rules, perimeter firewall policies, secure remote administration, centralized authentication, wireless security, and secure web communication.

Project Goals

- Design a structured and secure network architecture for the laboratory and partner clinics.
- Separate different classes of network traffic using VLANs.
- Restrict access to protected diagnostic services using ACLs.
- Secure network-device administration using SSH.
- Apply perimeter firewall policies and network address translation.
- Provide centralized wireless authentication using WPA2-Enterprise and Active Directory services.
- Protect web-based diagnostic communication using HTTPS/TLS.
- Validate network behaviour using traffic analysis.
- Automate basic clinic-node connectivity monitoring with Python.

Project Scope

The final solution was a simulated network design rather than a production deployment. The academic scope focused on demonstrating network segmentation, access control, secure management, authentication, traffic protection, testing, and basic automation.

2. Network & Security Architecture

The simulated network used multiple logical segments so that different types of systems and traffic could be isolated. The report describes dedicated VLANs for laboratory systems, diagnostic/server systems, wireless users, management, and Active Directory services.

Security / Network Layer	Implementation
Segmentation	VLAN-based separation of laboratory, diagnostic/server, wireless, management, and AD traffic.
Access control	Extended ACLs used to control communication between segments and restrict protected resources.
Perimeter security	Firewall rules applied at the network boundary, with NAT/PAT for address translation.
Device management	SSH used for secure remote administration instead of Telnet.
Wireless security	WPA2-Enterprise with WLC and centralized authentication services.
Application security	HTTPS/TLS used for secure diagnostic report communication.
Visibility / validation	Wireshark used to inspect and validate network traffic.

Core Technologies

- Cisco Packet Tracer and Cisco IOS commands
- VLANs and inter-VLAN routing
- Extended Access Control Lists (ACLs)
- Firewall rules and NAT/PAT
- SSH
- WPA2-Enterprise and Wireless LAN Controller
- Active Directory Domain Services
- HTTPS/TLS
- Wireshark
- Python

3. My Individual Contribution

This was a four-member group project. My role was **Network Implementer & Simulation Lead**. My primary technical responsibilities were network topology design, VLAN segmentation, ACL configuration, firewall rules, and SSH setup in Cisco Packet Tracer.

- Designed the network topology with a team member.
- Worked on server, VLAN, and firewall configuration with teammates.
- Worked primarily on ACL and SSH configuration.
- Contributed to the project risk plan and communication plan.
- Contributed to report writing and final formatting.
- Used Cisco IOS commands and Packet Tracer to implement and troubleshoot the design.

Security-Focused Implementation

My main technical focus was implementing and validating network security controls. This included isolating diagnostic, administrative, and wireless traffic with VLANs; restricting access with ACLs; securing device management with SSH; and troubleshooting connectivity and policy behaviour in the simulated topology.

The project documentation also records development of a Python script from scratch to alert when a clinic node becomes unresponsive. The script was tested in a separate simplified Packet Tracer topology while the main group topology was still being completed.

Evidence to Include in the Repository

- Sanitized Cisco Packet Tracer (.pkt) simulation
- Network topology diagram
- Selected ACL, VLAN, SSH, firewall, and wireless configuration screenshots
- Python monitoring script
- Selected traffic-validation evidence

4. Troubleshooting & Problem Solving

A major learning component of the project was troubleshooting configuration problems in a multi-segment network. Several issues were not obvious from the logical design alone and required systematic tracing and testing.

Problem	What I learned / how it was resolved
VLANs could not reach the internet	The trunk configuration was incomplete. Trunk mode and related VLAN settings were verified and corrected.
ACL did not block the intended traffic	The rule order was incorrect. Cisco ACL processing is order-dependent, so the permit/deny sequence was reviewed and corrected.
Early ACL configuration blocked legitimate traffic	ACLs and rule ordering were reviewed and the ACL was rebuilt and tested incrementally.
Testing depended on an incomplete ground truth	A topology-independent Packet Tracer topology was created to test the Python monitoring script without the need for a physical network.

Problem-Solving Approach

- Break the topology into smaller sections instead of troubleshooting the whole network at once.
- Use Packet Tracer simulation mode to trace packet behaviour.
- Check Cisco documentation rather than relying on assumptions about default behaviour.
- Build and test ACL rules incrementally.
- Validate both allowed and denied traffic after a security rule is changed.

5. Testing & Validation

The project documentation records 18 security and functionality test cases. The final reported result was that all 18 test cases passed before submission.

- VLAN connectivity and traffic isolation
- Inter-VLAN communication
- ACL enforcement and unauthorized-access prevention
- SSH access control
- Telnet restriction
- Wireless authentication
- Active Directory authentication
- HTTPS/TLS communication
- Firewall traffic filtering
- NAT/PAT operation
- Python clinic-node connectivity monitoring
- Broadcast-domain isolation

Validation Philosophy

The testing approach was not limited to checking whether devices could communicate. The project also tested whether communication that should be blocked was actually denied. This distinction is important for network security because a configuration can appear functional while still allowing unintended traffic.

6. Lessons Learned

The project provided practical experience in applying security concepts that are often first encountered theoretically. The most significant technical lesson was that ACL rule ordering is part of the security policy itself: placing an allow rule before a deny rule can change the result of the entire policy.

I also learned that switch and routing behaviour should be verified rather than assumed. The trunking issue demonstrated how a small configuration detail can prevent communication across an otherwise logically correct topology.

Another lesson was the value of systematic testing. Testing each network segment independently before combining everything would have made it easier to isolate faults and reduce debugging time.

Project & Teamwork Lessons

- Use shared version control or a controlled shared workspace from the beginning.
- Maintain a common record of IP addresses and configuration changes.
- Set clear internal deadlines and responsibilities.
- Test components individually before integrating the complete topology.
- Use documentation and repeatable troubleshooting steps instead of guessing.

7. Limitations & Future Improvements

Limitations

- The network was implemented in Cisco Packet Tracer, so it is a simulation rather than a production deployment.
- The simulated design does not provide full production-grade redundancy or failover.
- Production-level VPN infrastructure was outside the final implementation scope.
- Packet Tracer provides more limited functionality than real network hardware and security appliances.
- Advanced monitoring and intrusion-detection capabilities were not fully implemented.

Future Improvements

- Implement OSPF for more scalable routing.
- Establish VPN tunnels between sites.
- Add advanced network monitoring and visualization.
- Introduce wireless intrusion-detection capabilities.
- Improve redundancy and fault tolerance.
- Expand automated security monitoring and alerting.

8. Portfolio Summary

This project demonstrates practical exposure to network security implementation, troubleshooting, and basic automation. Its strongest portfolio evidence is the combination of the working Packet Tracer simulation, network topology, security-configuration screenshots, testing evidence, and Python monitoring script.

Academic project disclaimer: This repository is intended for educational and portfolio purposes. The original work was completed as a group project; this summary highlights my individual technical contribution and does not claim sole authorship of the complete project.

Source basis: project documentation and individual contribution section of the original academic report. No student IDs, signatures, or other personal submission details are included in this portfolio summary.